

Maharashtra Public Service Commission Assessment

Subject Name :	Lecturer Environment Maharashtra Education Service Group B Higher and Technical Education Dept
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 4410092188470 Question Type : MCQ

_____ is considered a classic form of urban sprawl.

Options :

1. Scattered development
2. Monocentric development
3. Polycentric development
4. Clustered development

Question Number : 2 Question Id : 4410092188474 Question Type : MCQ

Which GIS function is essential for planning optimized waste collection routes in cities?

Options :

1. Overlay analysis
2. Interpolation
3. Buffer analysis
4. Network analysis

Question Number : 3 Question Id : 4410092188471 Question Type : MCQ

Why do vegetated areas typically yield positive Normalized Difference Vegetation Index (NDVI) values in remote sensing analysis?

Options :

1. Vegetated areas reflect less near-infrared than red light
2. Vegetated areas absorb all near-infrared and red light
3. Vegetated areas reflect more near-infrared than red light
4. Vegetated areas reflect equal amounts of near-infrared and red light

Question Number : 4 Question Id : 4410092188473 Question Type : MCQ

What is a primary limitation of using remote sensing data alone for sustainable water resource management?

Options :

1. It cannot monitor surface temperature variations
2. It cannot classify surface and subsurface land use types
3. It cannot measure subsurface water properties directly
4. It cannot map physical properties of soil salinity accurately

Question Number : 5 Question Id : 4410092188475 Question Type : MCQ

Why is it important to use SAR data from ERS and RADARSAT for mapping flood-inundated areas, especially in India?

Options :

1. They produce better color imagery than other satellites
2. They are effective even during bad weather conditions
3. They have lower operational costs than optical sensors
4. They provide longer historical flood records

Question Number : 6 Question Id : 4410092188454 Question Type : MCQ

What is the fate of air that descends at the subtropical high-pressure belt before becoming surface winds in the Northern Hemisphere?

Options :

1. It splits into northward and southward currents, forming westerlies and trade winds
2. It moves solely northward toward the pole as a single current and trade winds
3. It moves solely southward toward the equator as a single current and trade winds
4. It remains stationary in the high-pressure zone and varies with altitude

Question Number : 7 Question Id : 4410092188457 Question Type : MCQ

How does the Coriolis effect influence the direction of air flow around low-pressure systems in the southern hemisphere?

Options :

1. Air circulates anticlockwise following a spiral-like path
2. Air circulates clockwise following a spiral-like path
3. Air moves directly outward in a straight path
4. Air remains stationary due to balanced forces

Question Number : 8 Question Id : 4410092188460 Question Type : MCQ

In a mountain valley, which wind phenomenon is most likely observed during the daytime due to topographical influence?

Options :

1. Up-draft or up-mountain breeze
2. Coastal fog formation in valleys
3. Monsoon wind reversal
4. Down-draft or down-mountain breeze

Question Number : 9 Question Id : 4410092188462 Question Type : MCQ

What does the length of each line in a standard wind rose diagram infer?

Options :

1. Temperature variation in that particular direction
2. Average wind speed from that direction
3. Proportional frequency of wind from that direction
4. Concentration of pollutants in each direction

Question Number : 10 Question Id : 4410092188463 Question Type : MCQ

Which of them best describes the irrigation sources and predominant soil in the Upper Gangetic Plains region?

Options :

1. Drip irrigation and clayey soil
2. Rainfed fields and black cotton soil
3. Sprinkler systems and alluvial soil
4. Canal, tube-well, wells and sandy loam

Question Number : 11 Question Id : 4410092135172 Question Type : MCQ

Which core component in ICP MS is responsible for ionizing the sample atoms?

Options :

1. Detector
2. Plasma
3. Quadrupole
4. Nebuliser

Question Number : 12 Question Id : 4410092135174 Question Type : MCQ

What property of a nucleus allows it to be detected by the NMR spectroscopy?

Options :

1. Size
2. Mass
3. Spin
4. Charge

Question Number : 13 Question Id : 4410092135165 Question Type : MCQ

Which molecule binds to hemoglobin and reduces oxygen transport in humans trapped in a building fire?

Options :

1. Sulfur dioxide
2. Nitrogen dioxide
3. Carbon dioxide
4. Carbon monoxide

Question Number : 14 Question Id : 4410092135166 Question Type : MCQ

Which chemical class does peroxyacetyl nitrate (PAN) belong to in atmospheric chemistry?

Options :

1. Secondary pollutant
2. Volatile organic
3. Primary pollutant
4. Organochlorine

Question Number : 15 Question Id : 4410092135167 Question Type : MCQ

Which characteristic makes persistent organic pollutants (POPs) particularly concerning in aquatic food chains?

Options :

1. Biotransformation
2. Volatilization
3. Bioaccumulation
4. Biomagnification

Question Number : 16 Question Id : 4410092135151 Question Type : MCQ

Which process predominantly governs the chemical speciation of mercury in the troposphere?

Options :

1. Thermal decomposition by sunlight
2. Microbial transformation in air
3. Direct hydrolysis with water vapor
4. Photochemical oxidation by ozone

Question Number : 17 Question Id : 4410092135156 Question Type : MCQ

During which stage of the hydrological cycle does the removal of dissolved salts/ions of water typically occur?

Options :

1. Groundwater recharge
2. Precipitation
3. Evaporation
4. Runoff

Question Number : 18 Question Id : 4410092135157 Question Type : MCQ

Match the following soil organic matter (SOM) fractions with their primary ecological function:

Column 1	Column 2
A. Particulate organic matter (POM)	1. Rapid nutrient turnover
B. Microbial biomass carbon	2. Long-term carbon sequestration
C. Black carbon	3. Energy source for microorganisms
D. Dissolved organic carbon (DOC)	4. Mobile carbon pool influencing leaching

Options :

1. A-4, B-3, C-2, D-1
2. A-1, B-2, C-4, D-3
3. A-3, B-1, C-2, D-4
4. A-2, B-4, C-1, D-3

Question Number : 19 Question Id : 4410092135158 Question Type : MCQ

How would an increase in soil moisture typically impact rate of carbon mineralization in temperate soils?

Options :

1. It always suppresses mineralization regardless of amount
2. It causes a continuous linear increase in mineralization
3. It increases mineralization up to a threshold before becoming inhibitory
4. It results in immediate complete inhibition of mineralization

Question Number : 20 Question Id : 4410092135160 Question Type : MCQ

Which feature distinguishes neonicotinoid pesticides from other major pesticide classes in the terms of their primary target site?

Options :

1. They disrupt sodium channels in neurons.
2. They block chloride channels in insect membranes.
3. They act as agonists of nicotinic acetylcholine receptors in insects.
4. They inhibit acetylcholinesterase enzymes.

Question Number : 21 Question Id : 4410092135828 Question Type : MCQ

In a food web, which term refers to species that are only consumed but do not consume others?

Options :

1. Intermediate species
2. Top predators
3. Basal species
4. Decomposers

Question Number : 22 Question Id : 4410092135855 Question Type : MCQ

Nagaland has the highest number of _____ to conserve biological diversity.

Options :

1. Wildlife sanctuaries
2. National parks
3. Community reserves
4. Conservation reserves

Question Number : 23 Question Id : 4410092135836 Question Type : MCQ

Which ecological transformation is most likely to occur, if a previously distinct grassland region undergoes a significant increase in annual rainfall and temperature stability?

Options :

1. The region may develop into a tropical rain forest biome
2. The area will transform into a tundra region
3. It will gradually become a shrubland biome
4. It will expand into a desert ecosystem

Question Number : 24 Question Id : 4410092135837 Question Type : MCQ

Which biome is most likely to replace a temperate forest if annual precipitation decreases from moderate to low levels while the average temperature remains nearly constant?

Options :

1. Chaparral biome
2. Temperate grassland
3. Conifer forest
4. Desert biome

Question Number : 25 Question Id : 4410092135843 Question Type : MCQ

A population of deer increases rapidly in a forest until food becomes scarce, causing the growth to slow. What is primarily regulating this population?

Options :

1. Predation by wolves
2. Natural disasters like wildfires
3. Competition for limited resources
4. Disease outbreak

Question Number : 26 Question Id : 4410092135845 Question Type : MCQ

Which factor distinguishes a keystone species from other less-abundant species in a community?

Options :

1. It is always the least abundant species present in the community.
2. It interacts directly with all other species equally.
3. It only creates new habitats without affecting interactions.
4. It exerts influence that is not proportional to its population size.

Question Number : 27 Question Id : 4410092135830 Question Type : MCQ

Which scenario would most likely decrease the stability of a fen wetland ecosystem?

Options :

1. Seasonal flooding, increasing waterfowl populations
2. Dominance of Sphagnum restricting plant diversity
3. Prolonged blocked drainage and stagnant water
4. Disruption of water movement through mineral soils

Question Number : 28 Question Id : 4410092135834 Question Type : MCQ

What is the effect of Earth's rotation on the distribution of salinity across an estuary in the Northern Hemisphere?

Options :

1. Earth's rotation increases salinity only near the river mouth
2. It results in higher salinity on the right side of the estuary
3. Rotation of earth creates uniform salinity throughout the estuary
4. It causes higher salinity on the left side relative to river-ocean flow

Question Number : 29 Question Id : 4410092135844 Question Type : MCQ

Which of the following is a major ecological consequence of uncontrolled population explosion in urban areas?

Options :

1. Expansion of forest cover in city regions
2. Decreased waste generation overall
3. Improved biodiversity in urban settings
4. Increased competition for limited natural resources

Question Number : 30 Question Id : 4410092135846 Question Type : MCQ

What is the primary function of warning coloration (aposematism) in prey species?

Options :

1. It enables prey to escape by blending into the habitat.
2. It signals predators that the animal is toxic or unpalatable.
3. It ensures the prey mimics another non-toxic species.
4. It always confuses predators by sudden color changes.

Question Number : 31 Question Id : 4410092135847 Question Type : MCQ

Which characteristic best distinguishes microparasites from macroparasites based on their life cycle and host interaction?

Options :

1. Microparasites have long generation times and rarely multiply in hosts
2. Macroparasites have long generation times and multiply rapidly within hosts
3. Macroparasites have short generation times and multiply rapidly within hosts
4. Microparasites have short generation times and multiply rapidly within hosts

Question Number : 32 Question Id : 4410092135854 Question Type : MCQ

Why is the lung at high risk for damage from toxicants?

Options :

1. It only contacts airborne toxicants and drugs directly
2. It is exposed to both inhaled and circulating toxicants frequently
3. It is unaffected by systemic circulation of drugs frequently
4. Its detoxication ability is greater than liver or other part of the body

Question Number : 33 Question Id : 4410092134709 Question Type : MCQ

Which layer of the Earth's atmosphere contains most of the atmospheric heat responsible for weather processes?

Options :

1. Stratosphere
2. Mesosphere
3. Troposphere
4. Exosphere

Question Number : 34 Question Id : 4410092134711 Question Type : MCQ

World Environment Day is observed on 5 June. Which major climatic event normally begins in India during the same month?

Options :

1. Retreat of the Southwest Monsoon
2. Onset of the Northeast Monsoon
3. Onset of the Southwest Monsoon
4. Development of winter westerlies

Question Number : 35 Question Id : 4410092134716 Question Type : MCQ

Which process contributes directly to geochemical recycling of elements in the environment?

Options :

1. Plate boundary formation
2. Meteorite impact events
3. Mineral weathering and sediment transport
4. Volcanic crater erosion

Question Number : 36 Question Id : 4410092134719 Question Type : MCQ

Which statement explains why clay typically has lower hydraulic conductivity than sand and gravel?

Options :

1. Clay contains smaller, poorly connected pore spaces
2. Clay contains larger, well-connected pore spaces
3. Clay contains fewer mineral grains per unit volume
4. Clay contains greater amounts of dissolved minerals

Question Number : 37 Question Id : 4410092134723 Question Type : MCQ

Match the hazard type in List I with the corresponding example in List II.

List I	List II
P. Geological hazard	1. Industrial chemical leak
Q. Biological hazard	2. Earthquake
R. Meteorological hazard	3. Cyclone
S. Technological hazard	4. Epidemic outbreak

Options :

1. P-2, Q-4, R-3, S-1
2. P-2, Q-3, R-4, S-1
3. P-1, Q-4, R-2, S-3
4. P-4, Q-2, R-1, S-3

Question Number : 38 Question Id : 4410092134725 Question Type : MCQ

The horizontal movement of a tsunami wave is initiated primarily by _____.

Options :

1. Seafloor sediment transport
2. Atmospheric pressure changes
3. Potential energy from water uplift
4. Coastal wave refraction

Question Number : 39 Question Id : 4410092134726 Question Type : MCQ

Which condition can lead to hydrological drought even after a period of rainfall?

Options :

1. High groundwater recharge
2. Increased soil moisture retention
3. Enhanced streamflow generation
4. Delayed recovery of water storage

Question Number : 40 Question Id : 4410092134728 Question Type : MCQ

Which objective is central to disaster management across all phases of the disaster cycle?

Options :

1. Restoration of damaged infrastructure
2. Provision of emergency medical services
3. Reduction of disaster risk and vulnerability
4. Distribution of relief materials

Question Number : 41 Question Id : 4410092134729 Question Type : MCQ

Which activity is a disaster preparedness measure rather than a mitigation measure?

Options :

1. Enforcing seismic building codes
2. Conducting evacuation drills
3. Constructing retaining structures
4. Stabilizing vulnerable slopes

Question Number : 42 Question Id : 4410092134730 Question Type : MCQ

An event is reclassified from a district disaster to a state disaster when_____.

Options :

1. Economic losses exceed a threshold
2. Impacts extend beyond district response capability
3. Media coverage increases substantially
4. Industrial facilities are affected

Question Number : 43 Question Id : 4410092134747 Question Type : MCQ

Methane in a biogas plant is primarily produced through the decomposition of organic matter under_____.

Options :

1. Acidic conditions
2. Aerobic conditions
3. Sterile conditions
4. Anaerobic conditions

Question Number : 44 Question Id : 4410092134750 Question Type : MCQ

Energy can be recovered from liquid waste primarily through the production of _____.

Options :

1. Hydrogen
2. Coal gas
3. Producer gas
4. Biogas

Question Number : 45 Question Id : 4410092134748 Question Type : MCQ

Why is hydrogen often referred to as an energy carrier rather than a primary energy source?

Options :

1. It contains lower energy than fossil fuels
2. It must be produced using primary energy resources
3. It generates carbon dioxide during combustion
4. It exists only under high-pressure conditions

Question Number : 46 Question Id : 4410092134752 Question Type : MCQ

Which energy source has traditionally contributed the largest share to India's commercial energy consumption?

Options :

1. Natural gas
2. Nuclear power
3. Hydropower
4. Coal

Question Number : 47 Question Id : 4410092134758 Question Type : MCQ

Which renewable energy technology primarily converts the kinetic energy of a naturally moving fluid into electrical energy through the rotation of turbine blades?

Options :

1. Solar photovoltaic system
2. Geothermal power plant
3. Biomass power plant
4. Wind turbine

Question Number : 48 Question Id : 4410092134734 Question Type : MCQ

Which fossil fuel is generally expected to produce the highest carbon dioxide emissions per unit energy generated?

Options :

1. Natural gas
2. Petroleum
3. Bituminous coal
4. Liquefied petroleum gas

Question Number : 49 Question Id : 4410092134736 Question Type : MCQ

The stability of natural gas hydrates in marine sediments is primarily controlled by _____.

Options :

1. High temperature and low pressure
2. Low temperature and high pressure
3. High salinity and low pressure
4. Low salinity and high temperature

Question Number : 50 Question Id : 4410092134739 Question Type : MCQ

Which geological condition is most favorable for large-scale geothermal power generation?

Options :

1. Regions with high groundwater salinity
2. Locations with intense surface weathering
3. Basins with thick sediment accumulation
4. Areas with elevated geothermal gradients

Question Number : 51 Question Id : 4410092134740 Question Type : MCQ

Which material characteristic is essential for generating electrical current in a photovoltaic cell upon exposure to sunlight?

Options :

1. Controlled electronic band gap
2. Elevated thermal resistance
3. High structural rigidity
4. Enhanced corrosion resistance

Question Number : 52 Question Id : 4410092134749 Question Type : MCQ

A municipal solid waste stream contains large amounts of metals, glass, and construction debris. Why would this waste stream be less suitable for energy recovery through combustion?

Options :

1. Higher calorific value of the waste
2. Greater biodegradability of the waste
3. Higher methane yield from the waste
4. Lower combustible fraction of the waste

Question Number : 53 Question Id : 4410092134759 Question Type : MCQ

Which of the following is an anthropogenic source of air pollution?

Options :

1. Volcanic eruption releasing volcanic gases
2. Forest ecosystem emitting natural hydrocarbons
3. Motor vehicles emitting exhaust pollutants
4. Sea spray generating airborne salt particles

Question Number : 54 Question Id : 4410092134766 Question Type : MCQ

Which effect of water pollution directly impacts aquatic animals?

Options :

1. Reduction in oxygen available for respiration
2. Increase in dissolved oxygen availability
3. Improvement in aquatic habitat quality
4. Enhancement of fish reproductive success

Question Number : 55 Question Id : 4410092134769 Question Type : MCQ

Which water quality parameter indicates the amount of oxygen required to chemically oxidize organic and inorganic pollutants in water?

Options :

1. Dissolved Oxygen (DO)
2. Salinity
3. Turbidity
4. Chemical Oxygen Demand (COD)

Question Number : 56 Question Id : 4410092134777 Question Type : MCQ

Which feature gives Advanced Oxidation Processes (AOPs) an advantage over conventional oxidation methods?

Options :

1. Generation of highly reactive hydroxyl radicals
2. Generation of highly reactive chlorine radicals
3. Generation of highly reactive sulfate radicals
4. Generation of highly reactive nitrate radicals

Question Number : 57 Question Id : 4410092134780 Question Type : MCQ

Which statement regarding industrial effluents and soil contamination is INCORRECT?

Options :

1. Heavy metals from effluents may accumulate in agricultural soils
2. Industrial effluents can alter the physicochemical properties of soil
3. Industrial effluents always improve soil fertility and crop productivity
4. Long-term effluent disposal may affect groundwater quality indirectly

Question Number : 58 Question Id : 4410092134791 Question Type : MCQ

What is the primary objective of incorporating a selector reactor in activated sludge bioreactor design?

Options :

1. To promote the growth of filamentous bacteria for sludge expansion
2. To favor floc-forming bacteria and improve sludge settling
3. To maximize oxygen transfer by increasing aeration intensity
4. To reduce return activated sludge flow to the aeration tank

Question Number : 59 Question Id : 4410092134765 Question Type : MCQ

Which adverse health outcome recognized by the Central Pollution Control Board (CPCB) is associated with prolonged exposure to environmental noise even when hearing loss is not evident?

Options :

1. Enhanced auditory adaptation and stress tolerance
2. Improved sleep efficiency and hormonal regulation
3. Reduced blood pressure due to acoustic habituation
4. Increased cardiovascular risk through physiological stress

Question Number : 60 Question Id : 4410092134770 Question Type : MCQ

Which statement about Biological Oxygen Demand (BOD) is incorrect?

Options :

1. BOD reflects the biodegradable organic load in water
2. BOD depends on the activity of microorganisms
3. BOD includes oxygen used for chemical oxidation
4. BOD is commonly used to assess organic pollution

Question Number : 61 Question Id : 4410092134774 Question Type : MCQ

Which statement about reverse osmosis for radionuclide removal is CORRECT?

Options :

1. It achieves high removal of both radium and uranium
2. It removes uranium more effectively than suspended solids only
3. It is effective only for radium removal from groundwater
4. It requires biological oxidation before radionuclide removal

Question Number : 62 Question Id : 4410092134776 Question Type : MCQ

Which statement correctly describes a characteristic of ion exchange resins used in wastewater treatment?

Options :

1. Strong-acid cation resins remain ionized across a wide pH range
2. Weak-acid cation resins remain ionized across a wide pH range
3. Weak-base anion resins remain ionized across a wide pH range
4. Chelating resins remain ionized across a wide pH range

Question Number : 63 Question Id : 4410092134783 Question Type : MCQ

Which impact of marine pollution poses the greatest risk to higher trophic levels through food-chain transfer?

Options :

1. Increase in tidal amplitude
2. Reduction in seawater density
3. Bioaccumulation of toxic contaminants
4. Increase in coastal sediment size

Question Number : 64 Question Id : 4410092134786 Question Type : MCQ

According to the Atomic Energy Regulatory Board (AERB), what is the annual effective dose limit for a member of the public under normal exposure conditions?

Options :

1. 0.1 mSv
2. 1 mSv
3. 10 mSv
4. 20 mSv

Question Number : 65 Question Id : 4410092134790 Question Type : MCQ

What happens to the alpha factor in a membrane bioreactor (MBR) when the Mixed Liquor Suspended Solids (MLSS) concentration increases from 8000 mg/L to 12,000 mg/L?

Options :

1. The alpha factor increases due to higher biomass concentration
2. The alpha factor decreases due to increased mixed liquor viscosity
3. The alpha factor remains constant regardless of MLSS concentration
4. The alpha factor becomes equal to one at higher MLSS concentration

Question Number : 66 Question Id : 4410092135059 Question Type : MCQ

What facility assists long-distance waste transportation?

Options :

1. Transfer station
2. Storage shed
3. Recycling bin
4. Compost yard

Question Number : 67 Question Id : 4410092135067 Question Type : MCQ

Which hazardous waste characteristic damages living tissues?

Options :

1. Physical size
2. Waste volume
3. Collection rate
4. Corrosive nature

Question Number : 68 Question Id : 4410092135058 Question Type : MCQ

Which advantage is associated with stationary collection systems?

Options :

1. Increased transfer station dependence
2. Reduced container transport requirements
3. Higher waste generation rates
4. Greater landfill excavation needs

Question Number : 69 Question Id : 4410092135068 Question Type : MCQ

Which exposure pathway commonly contributes to hazardous waste impacts?

Options :

1. Compost curing, moisture control
2. Landfill grading, cover placement
3. Collection, routing, vehicle allocation
4. Inhalation, ingestion, dermal absorption

Question Number : 70 Question Id : 4410092135071 Question Type : MCQ

Why is stabilization frequently applied before final disposal?

Options :

1. Improves long-term waste containment
2. Increases recycling market demand
3. Reduces collection crew requirements
4. Enhances biological decomposition rates

Question Number : 71 Question Id : 4410092135072 Question Type : MCQ

Which sources are most likely to generate pathological biomedical waste requiring specialized handling?

Options :

1. Residential kitchens, markets, and restaurants
2. Construction projects involving structural demolition
3. Hospitals, laboratories, clinics, and treatment facilities
4. Municipal parks, recreation areas, and sports complexes

Question Number : 72 Question Id : 4410092135074 Question Type : MCQ

When evaluating strategies for managing radioactive waste from nuclear power plants, which approach best minimizes long-term environmental risks?

Options :

1. Immediate recycling into commercial products
2. Near-surface disposal without engineered barriers
3. Long-term containment and environmental isolation
4. Frequent collection and rapid transportation

Question Number : 73 Question Id : 4410092135076 Question Type : MCQ

Which workplace factor most accurately represents a psychological occupational hazard?

Options :

1. Chronic stress from adverse work conditions
2. Excessive noise generated by equipment operation
3. Mechanical injuries from moving machinery
4. Dermal exposure to corrosive chemical agents

Question Number : 74 Question Id : 4410092135077 Question Type : MCQ

Which management objective is achieved through proper electronic waste recycling?

Options :

1. Resource recovery and pollution prevention simultaneously
2. Increased landfill dependence for final disposal
3. Elimination of all transportation requirements
4. Reduced collection productivity across municipalities

Question Number : 75 Question Id : 4410092135078 Question Type : MCQ

Which waste-generating activity contributes most significantly to plastic waste accumulation?

Options :

1. Atmospheric monitoring and weather forecasting
2. Geological mapping and resource exploration
3. Packaging and consumer product utilization
4. Hydrological assessment and watershed studies

Question Number : 76 Question Id : 4410092135089 Question Type : MCQ

Which forest category is recognized under the Indian Forest Act?

Options :

1. Coastal parks
2. Protected rivers
3. Reserved forests
4. Mining blocks

Question Number : 77 Question Id : 4410092135095 Question Type : MCQ

Which material recovery is promoted through the Batteries (Management and Handling) Rules, 2001?

Options :

1. Timber recovery
2. Sand recovery
3. Water recovery
4. Lead recovery

Question Number : 78 Question Id : 4410092135082 Question Type : MCQ

Which factors are commonly included during environmental cost-benefit evaluation?

Options :

1. Environmental audit attendance sheets
2. Forest species classification records
3. Industrial licensing documentation files
4. Monetary valuation of impacts

Question Number : 79 Question Id : 4410092135084 Question Type : MCQ

Eco-labeling schemes primarily encourage which market-based environmental approach?

Options :

1. Informed consumer purchasing decisions
2. Compulsory environmental litigation procedures
3. Government ownership of industries
4. Mandatory industrial pollution audits

Question Number : 80 Question Id : 4410092135094 Question Type : MCQ

Which concept forms the foundation of the E-Waste Management Rules, 2016?

Options :

1. Mandatory environmental litigation principle
2. Polluter immunity compensation principle
3. Resource extraction maximization principle
4. Extended producer responsibility principle

Question Number : 81 Question Id : 4410092135085 Question Type : MCQ

In environmental risk assessment, hazard identification primarily serves which purpose?

Options :

1. Conducting ecological restoration plans
2. Recognizing potential sources of harm
3. Estimating project financial returns
4. Preparing post-project audits

Question Number : 82 Question Id : 4410092135088 Question Type : MCQ

Under the Forest (Conservation) Act, 1980, which activity requires prior approval of the Central Government?

Options :

1. Preparation of forest inventory records
2. Diversion of forest land for non-forest use
3. Conduct of biodiversity awareness campaigns
4. Maintenance of village grazing registers

Question Number : 83 Question Id : 4410092135092 Question Type : MCQ

Which principle is most strongly reflected in the Hazardous and Other Wastes Rules, 2016?

Options :

1. Open disposal of industrial residues
2. Environmentally sound waste management
3. Voluntary waste tracking procedures
4. Unrestricted waste transportation policy

Question Number : 84 Question Id : 4410092135097 Question Type : MCQ

Which objective received the highest priority under the National Forest Policy, 1988?

Options :

1. Ecological stability and environmental balance
2. Commercial forestry export promotion
3. Expansion of forest-based industries
4. Industrial timber production enhancement

Question Number : 85 Question Id : 4410092135101 Question Type : MCQ

What is the principal objective of the United Nations Framework Convention on Climate Change (UNFCCC)?

Options :

1. Regulation of international carbon taxation
2. Stabilization of greenhouse gas concentrations
3. Elimination of all fossil fuels
4. Establishment of global energy markets

Question Number : 86 Question Id : 4410092135120 Question Type : MCQ

Which environmental application commonly uses forecasting techniques?

Options :

1. Species coding
2. Air quality
3. Plot numbering
4. Sample labeling

Question Number : 87 Question Id : 4410092135112 Question Type : MCQ

Which statistical procedures most commonly use the chi-square distribution as their test statistic?

Options :

1. Tests comparing population mean values
2. Models predicting outcomes using regression
3. Tests analyzing frequencies and goodness-of-fit
4. Methods estimating correlation coefficient values

Question Number : 88 Question Id : 4410092135114 Question Type : MCQ

Which assumption is commonly required for parametric t-testing?

Options :

1. Ordinal measurement scale
2. Perfect predictor independence
3. Approximately normal distribution
4. Equal category frequencies

Question Number : 89 Question Id : 4410092135118 Question Type : MCQ

What fundamental objective does environmental modelling primarily achieve within scientific decision-making processes?

Options :

1. Replacing empirical monitoring with theoretical assumptions entirely
2. Eliminating uncertainty from ecological observations and measurements
3. Representing environmental systems through simplified mathematical structures
4. Guaranteeing universally accurate predictions under changing conditions

Question Number : 90 Question Id : 4410092135119 Question Type : MCQ

What is the primary purpose of multiple regression models in environmental scientific analyses?

Options :

1. Relating one response variable to multiple predictors
2. Comparing categorical frequencies across independent observations
3. Estimating variance components from factorial experiments only
4. Evaluating temporal dependence within ecological populations exclusively

Question Number : 91 Question Id : 4410092135125 Question Type : MCQ

What is a major goal of the Himalayan Mission?

Options :

1. Forest Clearing
2. Urban Growth
3. Ecosystem Protection
4. Mining Expansion

Question Number : 92 Question Id : 4410092135139 Question Type : MCQ

Which factor directly strengthens national food security?

Options :

1. Forest Clearing
2. River Pollution
3. Sustainable Agriculture
4. Land Degradation

Question Number : 93 Question Id : 4410092135130 Question Type : MCQ

Why is rainwater harvesting promoted in water-stressed regions today?

Options :

1. Accelerates soil degradation processes
2. Encourages wastage of rainfall
3. Reduces pressure on aquifers
4. Increases surface runoff rates

Question Number : 94 Question Id : 4410092135138 Question Type : MCQ

Which condition best describes energy security within sustainability frameworks today?

Options :

1. Limited access to resources
2. Continuous energy demand growth
3. Exclusive dependence on imports
4. Reliable energy supply availability

Question Number : 95 Question Id : 4410092135148 Question Type : MCQ

Which lesson emerged strongly from the Bhopal Gas Disaster experience?

Options :

1. Advantages of hazardous storage
2. Benefits of reduced regulation
3. Need for increased emissions
4. Importance of industrial safety

Question Number : 96 Question Id : 4410092135124 Question Type : MCQ

Which policy outcome demonstrates successful implementation of Sustainable Habitat Mission objectives in cities?

Options :

1. Expanded construction over environmentally sensitive areas
2. Reduced energy demand through efficient infrastructure
3. Increased fossil dependence across urban sectors
4. Higher resource consumption without efficiency measures

Question Number : 97 Question Id : 4410092135131 Question Type : MCQ

Which watershed intervention most directly enhances groundwater recharge potential in semi-arid regions?

Options :

1. Accelerating runoff using concrete drainage networks
2. Increasing infiltration through recharge-oriented structures
3. Reducing vegetation cover across recharge zones
4. Encouraging surface water losses through channelization

Question Number : 98 Question Id : 4410092135133 Question Type : MCQ

Which measure contributes most directly to achieving Yamuna Action Plan pollution-control objectives?

Options :

1. Expanding sewage treatment and disposal facilities
2. Promoting unregulated waste dumping near banks
3. Increasing industrial discharge into river channels
4. Reducing wastewater treatment infrastructure investments

Question Number : 99 Question Id : 4410092135137 Question Type : MCQ

Why is adaptability considered essential for addressing uncertainties associated with future climate change?

Options :

1. Enhances resilience against changing environmental conditions
2. Prevents occurrence of all climate-related hazards
3. Guarantees complete elimination of greenhouse emissions
4. Ensures precise forecasting of climatic variations

Question Number : 100 Question Id : 4410092135145 Question Type : MCQ

Why are vehicular emission norms important for sustainable urban environmental management strategies?

Options :

1. Eliminating public transportation development initiatives entirely
2. Limiting transport-sector contributions to atmospheric pollution
3. Encouraging greater dependence on fossil-fuel vehicles
4. Increasing pollutant emissions from transportation networks